A High Performance NMOS-Switch High Swing Cascode Charge Pump for Phase-Locked Loops

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***Abstract*— In this work, we present a NMOS-switch high-swing cascode charge pump. The proposed design utilizes high-swing cascode current mirror to improve output current matching and a pull-up/pull-down mechanism to remedy the slow-node issue of a classic NMOS-switch current steering charge pump. The proposed NMOS-switch high-swing cascode charge pump has been designed using the IBM 0.13µm CMOS technology. Simulation results show that the proposed charge pump design features much improved output current matching and also eliminates the slow-node problem when compared to original approach.**

Index terms— Charge Pump, PLL, High-swing cascode structure

I. INTRODUCTION

The charge pump is a critical component of the Phase-Locked Loop (PLL) systems. The charge pump usually paired with a phase frequency detector (PFD) providing a wide locking range at a low cost. The function of the charge pump is to inject a constant amount of current into a loop filter. A simple charge pump consists of two switched current sources that pump current into or out of the loop filter according to the PFD outputs (Fig.1) [1].

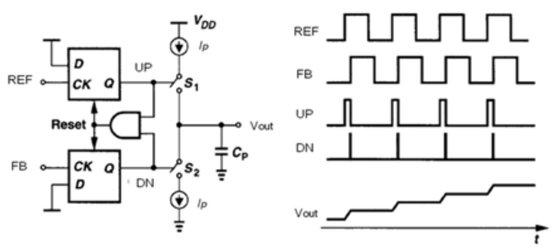

Figure 1. Charge pump model in PLL and related signal waveforms

The operation of the charge pump is described as follows. The two switches of the charge pump are controlled by the UP and DN signals from PFD. When the UP is high and DN is low, the current source of the charge pump becomes active and sources the current I_p into the load. When the UP is low and DN is high, the current sink becomes active and forces the current I_p to sink into the load. When both UP and DN are either high or low, no current should flow to or from the load. In this way, the output current of the charge pump is becoming proportional to the pulse width difference of UP and DN signals, and thereby, proportional to the the phase difference of the two inputs to PFD. A good charge pump should feature equal charge and discharge currents, minimum switching errors (such as charge injection, charge sharing, and clock feedthrough), and minimum output current mismatches.

In this paper, we propose a high performance NMOS-switch high-swing cascode charge pump based on a classic NMOS-switch current steering charge pump. The proposed charge pump aims at achieving a fast switching speed, high output current matching and alleviating the non-ideal effects existing in conventional charge pumps.

This paper is organized as follows. Section II describes non-ideal effects in conventional charge pumps. Section III discusses the classic NMOS-switch current steering charge pump and two disadvantages of the structure. Proposed NMOS-switch high-swing cascode charge pump and the simulation results are presented in Section IV. Section V provides conclusions and a summary.

II. NON-IDEAL EFFECTS IN A CONVENTIONAL CHARGE PUMP

The conventional charge pump is shown in Fig.2. In theory, an ideal charge-pump should provide a constant current immediately after one of its switches close. However, in a real phase-locked loop, the charge pump suffers from mismatches and current leakages [2]. These non-idealities could cause glitches on the VCO control voltage and lead to noisy PLL output. Therefore, the non-idealities of a charge pump should be carefully considered during the design.

Mismatches of MOS transistors are the leading cause for such non-ideal effects of a charge pump as *current mismatch* and *timing mismatch*. The current sources I_{UP} and I_{DN} are usually implemented using either current mirrors or appropriately biased transistors. A mismatch originates in the

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different type of MOS transistors used to implement the N-type and the P-type current sources. The resulting *current mismatch* occurs during charging and discharging the loop filter. Another type of mismatch is the *timing mismatch*. Since the UPB signal in Fig.2 is active low, therefore, an additional inverter is usually needed to produce the UPB signal from the UP signal of PFD. The extra delay of the inverter produces the *timing mismatch*.

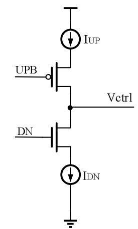

Figure 2. A conventional charge pump

Another non-ideal effect is the current leakage, which affects the voltage produced by the loop filter. *Charge injection, charge sharing* and *clock feedthrough* are the main sources of the leakage current. In the ON state, the switch transistors in Fig.2 hold charge in their channels. When one of the switches is turned OFF, a portion of the charge will flow to the load capacitance via its drain terminal, giving rise to *charge injection* error and resulting in the V_{ctrl} spike. The *Charge sharing* occurs when switches in charge pump are in the transition from OFF to ON state. The charge pump output voltage is floating when the switches are OFF. When the switches are turned ON, the *charge sharing* among the parasitic capacitances of the charge pump and the output node will occur, resulting in a deviation of the output voltage. The *Clock feedthrough* occurs when the fast rise and fall edges of a clock signal become coupled into the signal node via the gate-to-source and gate-to-drain overlap capacitances. The resulting rise in signal level could forward-bias the MOSFET's p-n junctions, leading to electron injection into the substrate that can potentially cause faulty operation, especially if it propagates through a nearby high-impedance node [3].

The output voltage V_{ctrl} glitches due to these non-ideal effects are indicated in Fig.3. Hence, a high performance charge pump should feature minimum current leakage, and zero mismatch between I_{UP} and I_{DN} current sources.

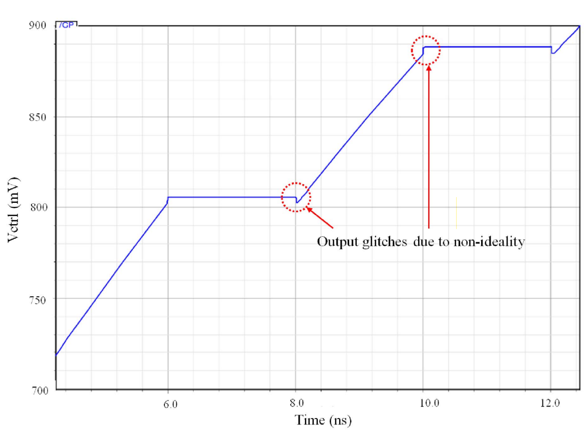

Figure 3. Conventional charge pump non-ideal effects

III. A CLASSIC NMOS-SWITCH CURRENT STEERING CHARGE PUMP

In order to improve the switching speed and minimize the switching mismatch issue of the conventional charge pumps, the classic NMOS-switches current steering charge pump has been proposed in [4] (Fig.4). It consists of two differential pairs (MN4 - MN5 and MN6 - MN7) that act as NMOS switches.

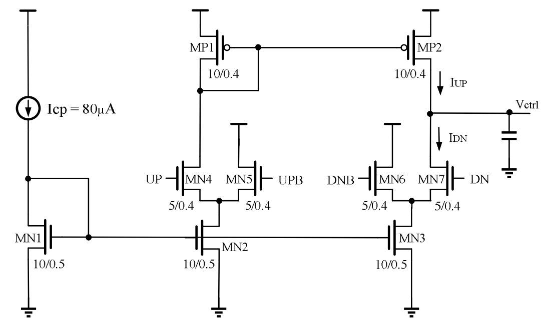

Figure 4. NMOS-switch current steering charge pump (W/L transistor dimensions in microns)

The NMOS-switch current-steering charge pump operates as follows. When the UP signal is high and DN signal is low, MN4 is turned ON, which turns on the PMOS current mirror (MP1-MP2). Hence, the charge pump current I_{UP} will flow in the MP2 and the output signal V_{ctrl} will be charged up. Meanwhile, the MN7 is turned OFF since the DN signal is low. The current of the MN3 current sink is steered into MN6. Similarly, when the UP signal is low and DN signal is high, the current I_{DN} is drawn through MN7 and the output signal V_{ctrl} is being discharged. When the UP and DN signals are driven high simultaneously, the current I_{UP} becomes equal to I_{DN}, resulting in the absence of charging or discharging of the output, thus keeping the output V_{ctrl} signal constant. Finally, when both UP and DN signals go low, both MN4 and MN7 will be turned OFF. All current mirrored from I_{cp} will be steered into MN5 and MN6. Therefore, no current will be flowing in or out of the output node to change the output voltage.

The NMOS-switch current steering charge pump has been designed for the pump current of 80μA. Fig.5 shows the process of charging up of the output voltage V_{ctrl} of the NMOS-switch current steering charge pump circuit. It can be seen that the output glitches due to the non-ideal effects have been reduced in comparison to the conventional charge pump reported in Fig.3. This charge pump, however, still suffers from two problems and would lead to output current noise. One of the problems is the presence of the slow-node in the circuit. The charge pump suffers from a slow-path node at the gate of MP2. During the transition from UP to DN, MN5 is switched ON, MN4 is switched OFF, and the current I_{UP} is expected to drop immediately to zero. However, MP1 still conducts current until its parasitic capacitance at the slow-node has been fully discharged, thus slowing down the decrease of I_{UP} to 0 (shown in Fig.5). This slow node problem also exists, however to a lesser degree, during the transition from DN to UP.

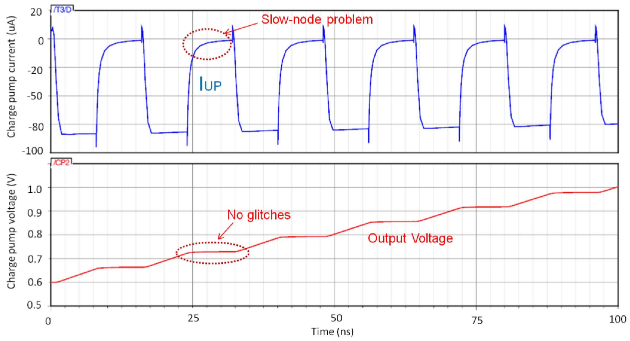

Figure 5. NMOS-switch current steering charge pump transient operation – (slow node issue)

Another problem of the design shown in Fig.4 is the large output current mismatch due to the limited output resistance (r_o) of the MN7 and MP2 transistors. This disadvantage may become even more significant in modern deep submicron CMOS technologies, since the output resistance of short channel transistors is smaller than that of long channel devices. The current mismatch resulted from moderate values of output resistances is well illustrated through comparison of charge pump output currents (I_{UP} and I_{DN}) (shown in Fig.6).

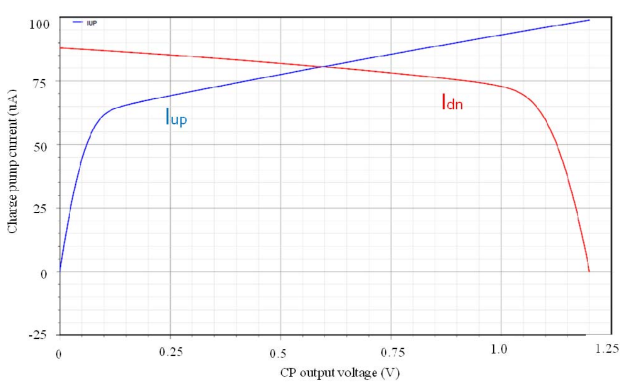

Figure 6. The output current mismatch of the NMOS-switch current steering charge pump in Fig.4 due to limited r_o

IV. DESIGN OF THE PROPOSED NMOS-SWITCH HIGH-SWING CASCODE CHARGE PUMP

An NMOS-switch high-swing cascode charge pump with improved output current matching and fast output discharging uses the architecture shown in Fig.7. In order to increase the output resistance of current mirrors of the output stage, two high-swing cascode current mirrors are used for the charge-up current I_{UP} and the discharge current I_{DN}. The charge-up current is provided by the p-channel high-swing cascode current mirror (MP1, MP2, MP3, MP4 and MP5), and the discharge current is provided by the n-channel high-swing cascode current mirror (MN1, MN8, MN9, MN10 and MN11). The output resistance of these cascode current mirrors is proportional to $g_m r_o^2$ instead of r_o in the original NMOS-switch charge pump shown in Fig.4. With the increased output resistance, the output current matching is significantly improved.

To solve the slow-node problem, a pull-up mirror [5] was used in the proposed charge pump design. The pull-up mirror for I_{UP} is formed by the MN4, MP8 and MP9 transistors. When the UP signal is low, the current in MP3 starts to drop to zero. Transistor MP8 mirrors the I_{cp} to MP9, pulling up the gate of MP3 to VDD so that MP3 could be turned off faster. This action remedies the slow-node problem and the cascode current source MP4 and MP5 can be quickly turned off. For the discharge current I_{DN}, the extra pull-down mechanism to remedy the slow node problem has been achieved by connecting the drain of MN7 to the gate of MN9.

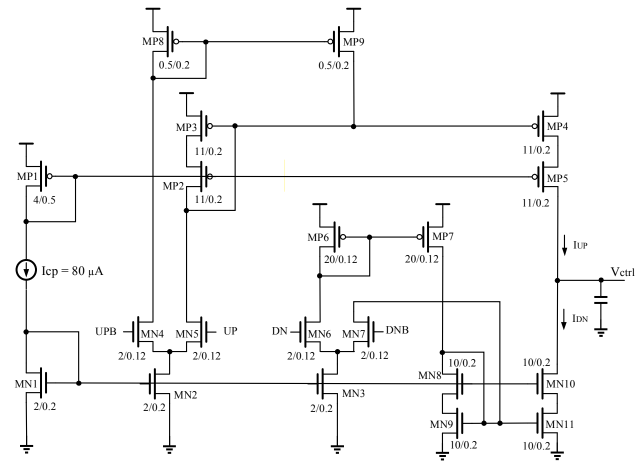

Figure 7. NMOS-switch high-swing cascode charge pump (W/L transistor dimensions in microns)

The proposed NMOS-switch high-swing cascode charge pump has been designed for a pump current of 80μA. The DC and transient operations of the design have been simulated to examine the charge pump performance. Fig.8 shows the DC operation of the charge pump, revealing the fact that the new charge pump architecture (Fig.7) significantly improves the output current matching (compared to Fig.6).

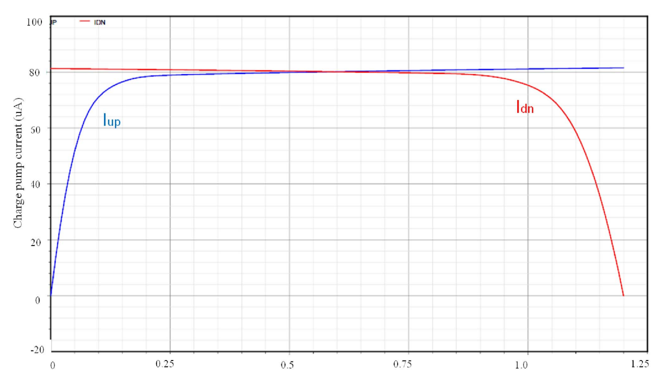

Figure 8. Output current matching of the NMOS-switch high-swing cascode charge pump

Fig.9 and Fig.10 show the process of charging and discharging of the output voltage V_{ctrl} of the NMOS-switch high-swing cascode charge pump circuit, respectively. It can be seen that the new charge pump circuit features a very fast charging and discharging process compared to the NMOS-switch current steering charge pump (Fig.5). Therefore, the new charge pump architecture has been found insensitive to the slow-node issues that deteriorate the operation of the original charge pump.

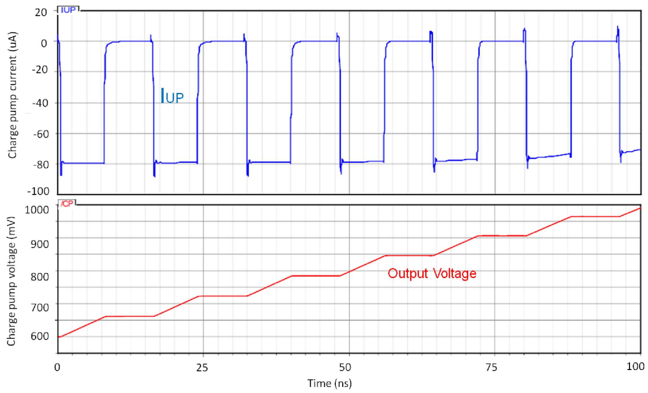

Figure 9. NMOS-switch high-swing cascode charge pump transient operation – charging phase (slow-node issue eliminated)

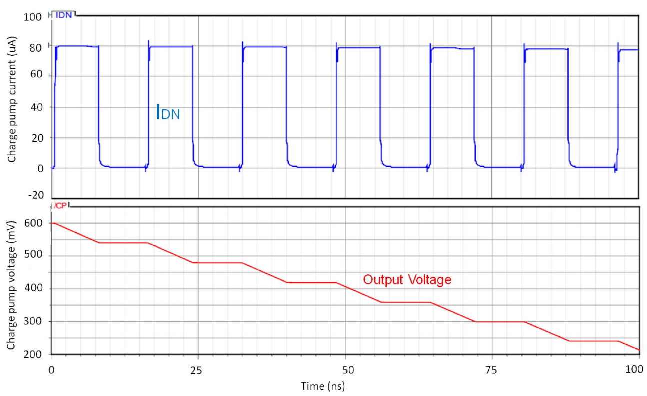

Figure 10. NMOS-switch high-swing cascode charge pump transient operation – discharing phase (slow-node issue eliminated)

To observe the superiority of the NMOS-switch high-swing cascode charge pump noise performance, both charge pumps (Fig.4 and Fig.7) have been connected to the output of the same PFD (Fig.1) and their noise characteristics have been simulated using SpectreRF noise analysis (Fig.11). The new NMOS-switch high-swing cascode charge pump features in 9dB improvement over the original charge pump in terms of current noise power spectral density. The excellent noise characteristics of the new charge pump make it an ideal component of the low-noise PLL systems [6].

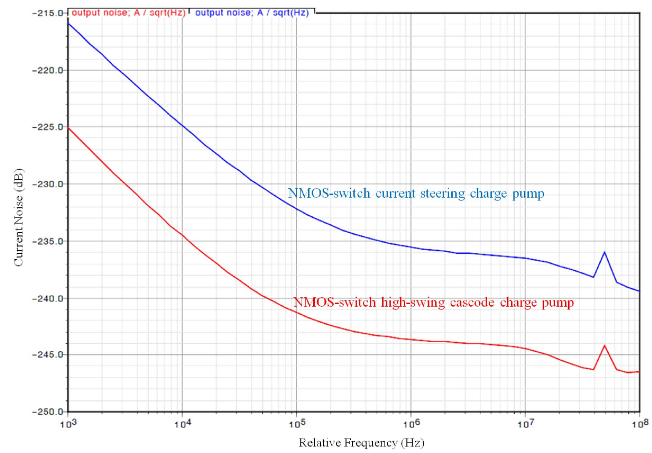

Figure 11. Current noise comparison of charge pumps

V. CONCLUSIONS

A high performance NMOS-switch high-swing cascode charge pump based on a classic NMOS-switch current steering charge pump has been designed and implemented using the IBM 0.13µm CMOS technology. Simulation results show that the new charge pump features improved current matching and eliminates the slow-node issue, leading to a further noise reduction. The proposed charge pump can be used in modern monolithic PLL design.

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